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Beverage container and carabiner assembly

Abstract

A beverage container and carabiner assembly including a container having a distal end and a proximal end, the container having a cavity therein, the container having a threaded mouth at the proximal end, the mouth in communication with the cavity, a threaded cap having a distal end and a proximal end, the cap having a through-bore therein proximate the proximal end, the cap operatively arranged to threadably and removably engage the threaded mouth of the container, and a carabiner, the carabiner having a proximal section, a distal section, and a clip, the distal section of the carabiner arranged within the through-bore of the cap.

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Background/Summary

FIELD

(1) The present invention relates to a beverage container having a carrying system removably secured in a cap of the container. More specifically, the invention relates to a carrying system which includes a carabiner secured within a cap of a water bottle.

BACKGROUND

(2) Reusable water bottles have become mainstream in recent years, primarily as an effort to reduce waste. Various name-brand companies have developed reusable water bottles with a plurality of features, such as, insulated walls to keep contents hot or cold and varieties of cap arrangements providing different drinking experiences.

(3) An issue with some water bottles relates to a means for carrying them. Due to their various sizes and shapes, carrying mechanisms have been developed in various configurations. One of these mechanisms is a cap having an integrated handle, typically constructed of the same material and extending therefrom.

(4) However, these handle solutions are arranged such that a closed loop is formed by the integrated handle, thereby requiring the use of another device to connect the handle to a desirable object, such as a backpack, travel bag, luggage, etc.

(5) In many arrangements, carabiner clips are used as a connecting element between the integrated handle and the object. Although this solution temporarily cures the abovementioned problem, it also introduces an additional component to the water bottle that can be removed and lost.

(6) An example of the aforementioned combination is disclosed in U.S. Published Patent Application No. 2021/0147127. The '127 application discloses a fluid container having a lid that is arranged to accept a carabiner therein. The bottle generally includes a vessel, or container having a lid and a magnet thereon. The container includes threads arranged within an inner surface of a mouth of the container that are arranged to removably secure the lid, by way of threads, to the container. The lid further includes a carabiner support with a substantially annular-ring shaped

configuration extending from an upper surface thereon, whereas the carabiner support is arranged to accept a carabiner therein. The carabiner can still be removed from the carabiner support.

(7) Therefore, it is desirable to create a handle that is more integrally arranged with the cap, i.e., no additional extending components, nor a carabiner that can be removed. U.S. Published Patent Application No. 2019/0152655 discloses a container and a cap having a handle integrated into the cap. The cap is disclosed to have a base with a wall arranged to have threading thereon. The base further includes oppositely disposed apertures. The base extends from a terminal end portion. The handle includes a sleeve with oppositely arranged end portions, whereas the end portions are arranged to be seated within apertures, allowing the handle to be rotatably connected to the base of the cap. The configuration does provide a handle but does not utilize a carabiner and is configured as a closed loop.

(8) Thus, there is a long felt need for a beverage container (e.g., a water bottle) having a cap, where the cap includes an integrated handle, where the handle takes the form of a carabiner and is integrated in a manner where the carabiner cannot be removed from the cap.

(9) There is also a long felt need for a carabiner and cap combination for a water bottle where the carabiner is integrated therein in such a way that no additional components extend from the cap.

SUMMARY

(10) According to aspects illustrated herein, the present invention generally comprises a beverage container and carabiner assembly including a container having a distal end and a proximal end, the container having a cavity therein, the container having a threaded mouth at the proximal end, the mouth in communication with the cavity, a threaded cap having a distal end and a proximal end, the cap having a through-bore therein proximate the proximal end, the cap operatively arranged to threadably and removably engage the threaded mouth of the container, and a carabiner, the carabiner having a proximal section, a distal section, and a clip, the distal section of the carabiner arranged within the through-bore of the cap.

(11) The present invention may also generally comprise a carabiner cap combination, the carabiner cap combination having a cap having a first end and a second end, a protrusion extending from the first end, the protrusion having threads arranged on an outer surface, a through-bore disposed therein and proximate the proximal end, and a carabiner, the carabiner having a proximal section, a distal section, and a clip, the distal section of the carabiner arranged within the through-bore of the cap, wherein the carabiner may rotate within the through-bore.

(12) In some embodiments, the present invention may comprise a carabiner cap combination including a cap having a proximal end and a distal end, a protrusion extending from the distal end, the protrusion having threads arranged on an outer surface, a through-bore disposed within the cap, and a carabiner, the carabiner having a proximal section, a distal section, and a clip, the distal section of the carabiner arranged for rotation within the through-bore of the cap.

(13) In other embodiments, the present invention may include a beverage container and carabiner assembly having a container having a cavity, the container having a threaded mouth in communication with the cavity, a threaded cap having a through-bore therein, the cap operatively arranged to threadably engage the threaded mouth of the container, and a carabiner, a part of which is fixedly secured within the through-bore of the cap.

(14) In some arrangements, the carabiner of the aforementioned present invention could comprise a frame, the frame having a nose arranged at a first end and a pivot end arranged at a second end, the frame having a spine arranged substantially between the nose and the pivot end, the frame including an arched portion arranged proximate the nose, the frame also including a longitudinal portion proximate the pivot end, and a gate pivotably secured to the pivot end and arranged to engage the nose, wherein the gate and the spine are spaced part at a first length, wherein the arched portion and the longitudinal portion are spaced apart at a second length, whereas the first length is greater than the second length.

(15) These and other objects, features, and advantages of the present invention will become readily

apparent upon a review of the following detailed description of the invention, in view of the drawings and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:
- (2) FIG. 1 is a perspective view of the present invention in use;
- (3) FIG. 2 is an exploded view of the present invention shown in FIG. 1;
- (4) FIG. 3 is a cross-sectional taken generally along line 3-3 in FIG. 1;
- (5) FIG. 4A is a front view of the cap of the present invention shown in FIG. 1;
- (6) FIG. 4B is a cross-sectional view taken generally along line 4B-4B in FIG. 4A;
- (7) FIGS. 5A through 5C are right-side views of the cap shown in FIG. 4;
- (8) FIGS. 6A through 6D are generally taken from DETAIL 6 shown in FIG. 3;
- (9) FIG. 7A is a perspective view of a carabiner of the present invention shown in FIG. 1;
- (10) FIG. 7B is a front view of the carabiner shown in FIG. 7A;
- (11) FIG. 8A is a perspective view of the cap shown of the present invention shown in FIG. 1 having an alternative embodiment of a carabiner therein;
- (12) FIG. 8B is a cross-sectional view taken generally along line 8B-8B in FIG. 8A;
- (13) FIG. 9A is a perspective view of the carabiner shown in FIG. 8A;
- (14) FIG. 9B is a front view of the carabiner shown in FIG. 8A;

DETAILED DESCRIPTION

- (15) At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.
- (16) Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to limit the scope of the claims.
- (17) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein can be used in the practice or testing of the example embodiments.
- (18) It should be appreciated that the term “substantially” is synonymous with terms such as “nearly,” “very nearly,” “about,” “approximately,” “around,” “bordering on,” “close to,” “essentially,” “in the neighborhood of,” “in the vicinity of,” etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term “proximate” is synonymous with terms such as “nearby,” “close,” “adjacent,” “neighboring,” “immediate,” “adjoining,” etc., and such terms may be used interchangeably as appearing in the specification and claims. The term “approximately” is intended to mean values within ten percent of the specified value.
- (19) It should be understood that use of “or” in the present application is with respect to a “non-exclusive” arrangement, unless stated otherwise. For example, when saying that “item x is A or B,” it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word “or” is not used to define an “exclusive or” arrangement. For example, an “exclusive or” arrangement for the statement “item x is A or B” would require that x can be only one of A and B. Furthermore, as used herein, “and/or” is intended

to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element.

(20) Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at least one of: a first element; a second element; and, a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element. A similar interpretation is intended when the phrase “used in at least one of:” or “at least one of” is used herein.

(21) Adverting now to the drawings. The following description should be taken in view of FIGS. 1 through 3. FIG. 1 illustrates the present invention, carabiner bottle 10 attached to article 2, specifically ring 4 of article 2. Carabiner bottle 10 includes carabiner 60 which allows bottle 10 to be removably secured to an object. FIGS. 2 and 3 generally illustrate an exploded view of carabiner bottle 10 and a cross-sectional view taken generally along line 3-3 in FIG. 1. Carabiner bottle 10 generally comprises container 12, carabiner cap combination 14, and O-ring 16.

(22) In some arrangements, container 12 includes distal end 18, proximal end 20, body 22, mouth 24, and neck 34. Body 22 has outer surface 26 and inner surface 28. Body 22 extends from distal end 18 proximate proximal end 20, thereby transitioning to neck 34. Neck 34 extends to proximal end 20 and is terminated by mouth 24. Mouth 24 is an aperture within neck 34. Threading 30 is arranged on inner surface 28 within mouth 24 and/or neck 34.

(23) Container 12 includes plurality of gripping members 32 arranged on body 22. There could also be a plurality of gripping members arranged on neck 34. Specifically, plurality of gripping members 32 extend from, or are within, or a combination thereof, outer surface 26 of container 12, arranged in either, or both, of body 22 and neck 34, preferably disposed between distal end 18 and proximal end 20. In a preferred embodiment, plurality of gripping members 32 are arranged on body 22 and disposed between distal end 18 and neck 34. Plurality of gripping members 32 is shown and discussed further in view of FIGS. 6A through 6F, *infra*.

(24) In some embodiments, container 12 is configured as a double wall insulated stainless steel container. In one embodiment, container 12 is configured as a single wall stainless steel container. It should be appreciated however, that alternative materials may be used in the construction of container 12, such as metal and plastic combinations, plastics, polymers, various different types of metals, etc. As such, those in the art will understand that any suitable material, now known or hereafter developed, may be used in forming container 10 described herein and *supra*.

(25) Carabiner bottle 10 also includes O-ring 16, which is arranged to be sandwiched between mouth 24 and/or proximal end 20 when carabiner cap combination 14 is engaged to container 12—thereby providing a water-tight seal between container 12 and carabiner cap combination 14.

(26) The following description should be taken in view of the aforementioned figures and FIGS. 4A and 4B. FIG. 4A generally illustrates a front view of carabiner cap combination 14 and FIG. 4B illustrates a cross-sectional view of same taken generally along line 4B-4B in FIG. 4A. Carabiner cap combination 14 is preferably a singular component having carabiner 60 fixedly secured within a through-bore disposed within body 15 of carabiner cap combination. In some embodiments, carabiner cap combination 14 comprises distal portion 36 and proximal portion 38, whereas distal

portion **36** and proximal portion **38** are arranged to be fixedly secured, having carabiner **60** therein, forming cap **14**.

(27) Distal portion **36** of cap **14** generally includes distal surface **40**, proximal surface **41**, outer surface **43**, protrusion **42**, and groove **46**. Protrusion **42** is arranged to extend from distal surface **40**. In a preferred embodiment, protrusion **42** takes a substantially cylindrical configuration that may also have a frustoconical portion arranged at an end of protrusion **42** arranged away from distal surface **40**. Threading **44** is arranged on an outside surface of protrusion **42**, whereas threading **44** is arranged to threadedly engage threading **30** of container **12**. Channel **48** is substantially arranged at least partially within distal surface **40** and preferably takes a circular configuration. Channel **48** is arranged to accept O-ring **16** therein, whereas O-ring **16** is substantially sandwiched within channel **48** when cap **14** is engaged to container **12**. O-ring **16** primarily increases the water-tight seal formed by threading **44** and threading **30**.

(28) In some embodiments, first cavity **45** and second cavity **47** are formed at least between proximal surface **41** and distal surface **40**, arranged between outer surface **43** and groove **46** (whereas groove **46** is positioned within groove body **46a**). In other arrangements, first cavity **45** and second cavity **47** may extend past distal surface **40** and into protrusion **42**.

(29) In some configurations, proximal portion **38** of cap **14** generally includes distal surface **50**, proximal surface **52**, groove **54**, first protruding section **56**, and second protruding section **58**. First protruding section **56** and second protruding section **58** are arranged to extend from distal surface **50**. In some embodiments, first protruding section **56** and second protruding section **58** may also include first cavity **55** and second cavity **57**, respectively. Groove **54** is arranged within groove body **54a**.

(30) Carabiner **60** generally includes distal section **62**, proximal section **64**, connection section **66**, and clip **68**. Distal section **62** at least includes a straight portion, i.e., a non-bent portion, that at least has a length which is the same as a length of both grooves **46** and **54**. In a preferred embodiment, distal section **62** has a length which is greater than a length of both grooves **46** and **54**. Distal section **62** extends from opposing ends into clip **68** and connection section **66**, whereas clip **68** and connection section **66** extend in a direction away from distal section. Proximal section **64** is joined to the respective ends of clip **68** and connection section **66** that extend away from distal section **62**. Carabiner **60** is generally configured to have a substantially circular cross-section.

(31) In some embodiments, carabiner **60** may include distal section **62**, proximal section **64**, connection section **66**, and clip **68**. Distal section **62** may include first end **62a**, second end **62b**, and pin **62c** disposed within an aperture arranged proximate first end **62a**. Distal section **62** may also include longitudinal portion **62d**, substantially disposed between first end **62a** and second end **62b**, whereas longitudinal portion **62d** includes longitudinal axis LX. Proximal section **64** may include first end **64a** and second end **64b**. Connecting section **66** may include first end **66a** and second end **66b**. Clip **68** may include first end **66a**, second end **66b**, and aperture **68c**. In some embodiments, clip **68** may also include locking sleeve **69**, which may be threadedly engaged to clip **68** such that rotation there moves locking sleeve in either a direction towards and past first end **66a** (i.e., a locked position) or a direction towards second end **66b** (i.e., an unlocked position). Second end **62b** of distal section **62** is connected to first end **66a** of connecting section **66**. Second end **66b** of connecting section **66** is connected to second end **64b** of proximal section **64**. First end **64a** of proximal section **64** and first end **68a** of clip **68** are arranged to have complimentary arrangements, such that first end **68a** of clip **68** abuts first end **64a** of proximal section, thereby restricting first end **68a** from moving in direction D1, i.e., past first end **64a** (but can move in direction D2). Clip **68** is rotatably secured to distal section **62** via pin **62c** within an aperture of first end **62a** of distal section **62** and within aperture **68c** of clip **68**. In a preferred embodiment, and well known within the art, clip **68** includes a spring mechanism such that first end **68a** of clip **68** maintains contact with first end **64a** of proximal section **64** until force, generally in direction D2, is imparted on clip **68**.

(32) When cap **14** is assembled, body **15** of cap **14** is formed by proximal portion **38** (bounded by distal surface **50** and proximal surface **52**) and a portion of distal portion **36** (the portion bounded by proximal surface **41** and distal surface **40**), thus, it can be said that body **15** of cap **14** is the portion of cap **14** substantially arranged between proximal surface **52** and distal surface **40**, best shown in FIG. **4B**. When cap **14** is assembled, first protruding section **56** and second protruding section **58** of proximal portion **38** are arranged to be at least partially seated within cavities **45** and **47** of distal portion **36**, whereas groove **54** and groove **46** thereby form the through-bore of cap **14** which secured holds carabiner **60** therein. In a preferred embodiment, the through-bore (comprised of groove **54** and groove **46**) has an internal circumference that is less than an outer circumference of distal portion **62** of carabiner **60**—allowing carabiner **60** to rotate within, shown in FIGS. **5A** through **5C**.

(33) The following description should be taken in view of the aforementioned figures and FIGS. **5A** through **5C** which generally illustrate a side view of carabiner cap combination **14**. As illustrated and discussed supra, carabiner **60** is fixedly secured within the through-bore (comprised of groove **54** and groove **46**) of cap **14** such that carabiner may rotate about axis **AX** of distal section **62** of carabiner **60**, shown representatively in FIGS. **5B** and **5C** specifically.

(34) The following description should be taken in view of FIG. **2** and FIGS. **6A** through **6D**. FIGS. **6A** through **6D** illustrate an enlarged portion of container **12** taken generally from DETAIL **6** in FIG. **2**. As discussed supra, container **12** may include plurality of gripping members **32** disposed on, within, or a combination thereof, outer surface **26**. Each of plurality of gripping members **32** may be comprised of at least one of: sinusoidal groove **70** (shown in FIG. **6A**); sinusoidal protrusion **72** (shown in FIG. **6A**); annular groove **78** (shown in FIG. **6B**); annular protrusion **80** (shown in FIG. **6B**); dome-like protrusion **74** (shown in FIG. **6C**); and, dome-like indentation **76** (shown in FIG. **6D**). As such, plurality of gripping members **32** may be comprised of one of the aforementioned configurations or a combination of the aforementioned configurations, e.g., plurality of gripping members **32** may comprise a plurality of sinusoidal grooves **70** and a plurality of sinusoidal protrusions **72** (as shown in FIG. **6A**), or plurality of gripping members **32** may comprise a plurality of annular grooves **78** and a plurality of annular protrusions **80** (as shown in FIG. **6B**), or other combinations. As such, various arrangements of the plurality of gripping members, in addition to the coverage of the members on the outer surface of the container (i.e., partially covered, completely cover, etc.) are within the scope of the appended claims.

(35) The following description should be taken in view of FIGS. **7A** and **7B** which generally illustrates an embodiment of a carabiner used in some embodiments of the present invention. In some embodiments, carabiner **100** generally includes frame **102** having spine **104** disposed opposite of gate **112**. Frame **102** is terminated by two end sections, nose **106** and pivot end **108**. Pivot end **108** includes an aperture therein, where the aperture is arranged to accept pivot pin **110**, thereby rotatably securing gate **112** to pivot end **108**. In some arrangements, frame **102** includes elongated curve portion **114** which is arranged opposite of longitudinal portion **116**, whereas spine **104** connects elongated curve portion **114** and longitudinal portion **116**. Longitudinal portion **116** also includes longitudinal axis **LX**. Carabiner **100** is generally configured to have a substantially circular cross-section.

(36) Pivot pin **110** could be a rivet, a fixed secured pin, or other like structure, so long as it can rotatably secure gate **112** to pivot end **108**.

(37) In a preferred embodiment, gate **112** includes a spring mechanism, or like mechanism, such that gate **112** forcibly contacts nose **106** until force is imparted thereon—allowing gate **112** to move in a direction towards spine **104**. In some embodiments, gate **112** could also include a lock, such as a screw-lock which would threadably secure to nose **106**, or a twist lock allowing gate **112** to be turned a specific degree to allow gate **112** to be released from nose **106**, or a twist and pull lock which requires gate **112** to be twisted and slid down to release gate **112** from nose **106**.

(38) In some embodiments, frame **102** also includes pivot portion **118** which includes pivot end

108 and major arch **120** having angle **A1**. Pivot portion **118** is the connecting section between pivot end **108** and longitudinal portion **116**. Longitudinal portion **116** includes inner face, or surface, **140**. Longitudinal portion **116** and spine **104** are connected by first curved portion **122** having major arch **124**. Major arch **124** includes angle **A2**, whereas angle **A1** and angle **A2** are substantially equal. Spine **104** includes inner face, or side, **138**. Spine **104** and elongated curve portion **114** are connected by second curved portion **126** having minor arch **128**, whereas minor arch **128** has angle **A3**. Elongated curve portion **114** includes elongated arch **130** having angle **A4**. Nose portion **132** includes nose **106** and is connected to elongated curve portion **114**. Nose portion **132** includes minor arch **134** having angle **A5**. Gate **112** also includes inner face **136**.

(39) The distance between inner face **136** of gate **112** (when gate **112** is abutting nose **106**) and inner face **138** of spine **104** is length **L1**. The distance between inner face, or surface **140** of longitudinal portion **116** and elongated arch **130** of elongated curve portion **114** is length **L2**. Length **L1** is greater than length **L2**. In a preferred embodiment, length **L1** and length **L2** are in an approximate ratio of five to three, respectively. In other embodiments, length **L2** may be approximately 50%-75% of length **L1**. In a preferred embodiment, length **L1** is at least 66% longer than length **L1**.

(40) Angle **A1** of major arch **120** is substantially equal to angle **A2** of major arch **124**. Angle **A3** of minor arch **128** is substantially equal to angle **A5** of minor arch **134**. Angles **A1** and **A2** are less than angles **A3** and **A5**. Angle **A4** of elongated arch **130** is greater than angles **A3** and **A5**. Angle **A4** of elongated arch **130** is greater than angles **A1** and **A2**.

(41) It should be noted that gate **112** is substantially equivalent to clip **68**. Longitudinal portion **116** is substantially equivalent to distal section **62**. Spine **104** is substantially equivalent to connecting section **66**. Elongated curve portion **114** is substantially equivalent to proximal section **64**. Nose **106** is substantially equivalent to first end **64a**. Pivot end **108** is substantially equivalent to first end **62a**. Pivot pin **110** is substantially equivalent to pivot pin **110**. It also should be appreciated that either carabiner **60** or carabiner **100** may be used in cap **14** of the present invention.

(42) The following description should be taken in view of the aforementioned figures and their respective description and FIG. **8A** through **9B**. FIG. **8A** through **9B** generally illustrate a perspective view and a cross-sectional view of an alternative embodiment of carabiner cap combination **14**, carabiner cap combination **200**, having an alternative embodiment of carabiners **60** and **100**, carabiner **250**, therein, and a perspective view and a front view of carabiner **250**.

(43) Carabiner cap combination **200** is preferably a singular component having carabiner **250** (or carabiner **60** or carabiner **100**) fixedly secured within a through-bore therein. In some embodiments, carabiner cap combination **200** comprises distal portion **202** and proximal portion **204**, whereas distal portion **202** and proximal portion **204** are arranged to be fixedly secured, having carabiner **250** therein, forming Carabiner cap combination **200**.

(44) In some embodiments, distal portion **202** of cap **200** generally includes distal surface **212**, proximal surface **213**, external surface **210**, internal surface **211**, and groove **206**. In some arrangements, external surface **210**, proximate distal surface **212** may have threading (similar to threading **44**) arranged thereon. In other arrangements, internal surface **211**, proximate distal surface **212** and within cavity **214**, may have threading arranged thereon (similar to threading **44**)—this arrangement may be configured to correspond to threading **30** of container **12** which may be alternatively arranged on outer surface **26** of mouth **24** of container **12**. O-ring channel **216** is arranged within cavity **214** proximate groove **206** and is arranged to accept O-ring **218** therein. Groove **206** includes first outer portion **206a**, second outer portion **206b**, and inner portion **206c**, where first outer portion **206a** and second outer portion **206b** have a great diameter than inner portion **206c**.

(45) In some arrangements, proximal portion **204** of carabiner cap combination **200** generally includes distal surface **220**, proximal surface **222**, and groove **208**. Plurality of ribs **224** extend from distal surface **220** into groove **208** (i.e., protruding into groove **208**). Groove **208** includes

first outer portion **208a**, second outer portion **208b**, and inner portion **208c**, where first outer portion **208a** and second outer portion **208b** have a diameter greater than the diameter of inner portion **208c**.

(46) Grooves **206** and **208**, when distal portion **202** and proximal portion **204** are joined, form the through-bore that fixedly and rotatably holds carabiner **250**, specifically longitudinal portion **266**, therein.

(47) In some embodiments, carabiner **250** generally includes frame **252** having spine **254** disposed opposite of gate **262**. Frame **252** is terminated by two end sections, nose **256** and pivot end **258**. Pivot end **258** includes an aperture therein, where the aperture is arranged to accept pivot pin **260**, thereby rotatably securing gate **262** to pivot end **258**. In some arrangements, frame **252** includes elongated curve portion **264** which is arranged opposite of longitudinal portion **266**, whereas spine **254** connects elongated curve portion **264** and longitudinal portion **266**. Longitudinal portion **266** includes first outer section **266a**, second outer section **266b**, and inner section **266c**, whereas first outer section **266a** and second outer section **266b** have a greater diameter than inner section **266c**. Longitudinal portion **266** also includes first groove **266d** disposed within first outer section **266a** and second groove **266e** disposed within second outer section **266b**. First groove **266d** and second groove **266e** are annular channels and are arranged to accept first carabiner O-ring **292** and second carabiner O-ring **294** therein, respectively. Longitudinal portion **266** also includes longitudinal axis LX. Longitudinal portion **266** is arranged to have a substantially circular cross-section.

(48) Pivot pin **260** could be a rivet, a fixed secured pin, or other like structure, so long as it can rotatably secure gate **262** to pivot end **258**.

(49) In a preferred embodiment, gate **262** includes a spring mechanism, or like mechanism, such that gate **262** forcibly contacts nose **256** until force is imparted thereon—allowing gate **262** to move in a direction towards spine **254**. In some embodiments, gate **262** could also include a lock, such as a screw-lock which would threadably secure to nose **256**, or a twist lock allowing gate **262** to be turned a specific degree to allow gate **262** to be released from nose **256**, or a twist and pull lock which requires gate **262** to be twisted and slid down to release gate **262** from nose **256**.

(50) In some embodiments, frame **252** also includes pivot portion **268** which includes pivot end **258** and major arch **270** having angle A6. Pivot portion **268** is the connecting section between pivot end **258** and longitudinal portion **266**. Longitudinal portion **266** includes inner faces, or surfaces, **290a**, **290b**, and **290c**, arranged on first outer section **266a**, second outer section **266b**, and inner section **266c**, respectively. Longitudinal portion **266** (specifically, second outer section **266b**) and spine **254** are connected by first curved portion **272** having major arch **274**. Major arch **274** includes angle A7, whereas angle A6 and angle A7 are substantially equal. Spine **254** includes inner face, or side, **288**. Spine **254** and elongated curve portion **264** are connected by second curved portion **276** having minor arch **278**, whereas minor arch **278** has angle A8. Elongated curve portion **264** includes elongated arch **280** having angle A9. Nose portion **282** includes nose **256** and is connected to elongated curve portion **264**. Nose portion **282** includes minor arch **284** having angle A10, whereas angle A10 and A8 are substantially equal. Gate **262** also includes inner face **286**.

(51) The distance between inner face **286** of gate **262** (when gate **262** is abutting nose **256**) and inner face **288** of spine **254** is length L3. The distance between inner faces, or surfaces **290a** and **290b** of first outer section **266a** and second outer section **266b** of longitudinal portion **266** and elongated arch **280** of elongated curve portion **264** is length L4. Length L3 is greater than length L4. In a preferred embodiment, length L3 and length L4 are in an approximate ratio of five to three, respectively. In other embodiments, length L4 may be approximately 50%-75% of length L3. In a preferred embodiment, length L3 is at least 66% longer than length L4.

(52) Angle A6 of major arch **270** is substantially equal to angle A7 of major arch **274**. Angle A8 of minor arch **278** is substantially equal to angle A10 of minor arch **284**. Angles A6 and A7 are less than angles A8 and A10. Angle A9 of elongated arch **280** is greater than angles A8 and A10, and also may be greater than angle A4 of carabiner **100** (shown in FIG. 7B). Angle A9 of elongated

arch **280** is greater than angles **A6** and **A7**. In some embodiments, angles **A8** and **A10** may be less than angles **A3** and **A5** of carabiner **100** (shown in FIG. 7B). The relationships between the angles of carabiner **100** and carabiner **250**, are recited below:

(53) $A1 = A2A3 = A5A1 < A3A1 < A5A2 < A3A2 < A5A1 < A4A2 < A4A3 < A4A6 = A7$
 $A8 = A10A6 < A8A6 < A10A7 < A8A7 < A10A6 < A9A7 < A9A8 < A9A10 < A9A4 < A9$
 $A1 = A6A1 = A7A2 = A6A2 = A7A8 < A3A8 < A5A9 < A3A9 < A5$

(54) It also should be appreciated that either carabiner **60**, carabiner **100**, or carabiner **200** may be used in cap **14** of the present invention.

(55) It should be appreciated that various aspects of the disclosure above and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

REFERENCE NUMERALS

(56) **2** Article **4** Ring **10** Carabiner bottle **12** Container **14** Carabiner cap combination **15** Body of cap **14** **16** O-Ring **18** Distal end **20** Proximal end **22** Body **24** Mouth **26** Outer surface **28** Inner surface **30** Threading **32** Plurality of gripping members **34** Neck **36** Distal portion **38** Proximal portion **40** Distal surface **41** Proximal surface **42** Protrusion **43** Outer surface **44** Threading **45** First cavity **46** Groove **46a** Groove body **47** Second cavity **48** Channel **50** Distal surface **52** Proximal surface **54** Groove **54a** Groove body **55** First cavity **56** First protruding section **57** Second cavity **58** Second protruding section **60** Carabiner **62** Distal section **62a** First end **62b** Second end **62c** Pin **62d** Longitudinal portion **64** Proximal section **64a** First end **64b** Second end **66** Connecting section **66a** First end **66b** Second end **68** Clip **68a** First end **68b** Second end **68c** Aperture **69** Locking sleeve **70** Sinusoidal groove **72** Sinusoidal protrusion **74** Dome-like protrusions **76** Dome-like indentations **78** Annular groove **80** Annular protrusion **100** Carabiner **102** Frame **104** Spine **106** Nose **108** Pivot end **110** Pivot pin **112** Gate **114** Elongated curve portion **116** Longitudinal portion **118** Pivot portion **120** Major arch **122** First curved portion **124** Major arch **126** Second curved portion **128** Minor arch **130** Elongated arch **132** Nose portion **134** Minor arch **136** Inner face **138** Inner face **140** Inner face **200** Cap **202** Distal portion **204** Proximal portion **206** Groove **206a** First outer portion **206b** Second outer portion **206c** Inner portion **208** Groove **208a** First outer portion **208b** Second outer portion **208c** Inner portion **210** External surface **211** Internal surface **212** Distal surface **213** Proximal surface **214** Cavity **216** O-ring channel **218** O-ring **220** Distal surface **222** Proximal surface **224** Plurality of ribs **250** Carabiner **252** Frame **254** Spine **256** Nose **258** Pivot end **260** Pivot pin **262** Gate **264** Elongated curve portion **266** Longitudinal portion **266a** First outer section **266b** Second outer section **266c** Inner section **266d** First groove **266e** Second groove **268** Pivot portion **270** Major arch **272** First curved portion **274** Major arch **276** Second curved portion **278** Minor arch **280** Elongated arch **282** Nose portion **284** Minor arch **286** Inner face **288** Inner face **290a** Inner face **290b** Inner face **290c** Inner face **292** First carabiner O-ring **294** Second carabiner O-ring **A1** First angle **A2** Second angle **A3** Third angle **A4** Fourth angle **A5** Fifth angle **A6** Sixth angle **A7** Seventh angle **A8** Eighth angle **A9** Ninth angle **A10** Tenth angle **AX** **D1** Direction **D2** Direction **D3** Direction **D4** Direction **L1** First length **L2** Second length **L3** Third length **L4** Fourth length **LX** Longitudinal axis

Claims

1. A carabiner cap combination, comprising: a cap having a proximal end and a distal end, a protrusion extending from said distal end, said protrusion having threads arranged on an outer surface, said cap having a distal portion having a lower groove arranged therein, said cap having a proximal portion having an upper groove arranged therein, wherein said proximal portion is arranged to be fixedly secured to said distal portion such that said lower groove and said upper

groove form a through-bore within said cap and between said proximal and distal ends of said cap, said through-bore having a longitudinal axis and a pair of openings disposed colinearly about said longitudinal axis, said pair of openings arranged on an external surface of said cap; and, a carabiner, said carabiner having a proximal section, a distal section, and a gate, said distal section of said carabiner fixedly and directly secured to said cap within said through-bore, wherein said carabiner is adapted to rotate within said through-bore of said cap.

2. A container having a proximal end and a distal end, said container comprising a mouth arranged at said proximal end, wherein said protrusion of said carabiner cap combination recited in claim 1 is arranged to be threadedly secured to said mouth of said container, thereby forming a container and carabiner cap combination.

3. The container recited in claim 2 further comprising an outer surface and a plurality of gripping members arranged on and circumscribing at least a portion of said outer surface of said container.

4. The container recited in claim 3, wherein said plurality of gripping members are arranged on a portion of said outer surface and disposed between said first end and said second end of said container.

5. The container recited in claim 4, where each of said plurality of gripping members may comprises at least one of: a plurality of sinusoidal grooves; a plurality of sinusoidal protrusions; a plurality of annular grooves; a plurality of annular protrusions; a plurality of dome-like indentations; and, a plurality of dome-like protrusions.

6. The container and carabiner cap combination recited in claim 2 further comprising an O-ring arranged to be sandwiched between said mouth of said container and said protrusion of said cap when said cap is engaged to said mouth.

7. The cabined cap combination recited in claim 1, wherein said carabiner comprises: a frame comprised of said proximal section of said carabiner and aid distal section of said carabiner, said frame having a nose arranged at a first end and a pivot end arranged at a second end, said frame having a spine arranged substantially between said nose and said pivot end, said frame including an arched portion arranged proximate said nose, said frame also including a longitudinal portion proximate said pivot end, said longitudinal portion being a part of the distal section of said carabiner; and, said gate pivotably secured to said pivot end and arranged to engage said nose, wherein said gate and said spine are spaced part at a first length, wherein said arched portion and said longitudinal portion are spaced apart at a second length, whereas said first length is greater than said second length, wherein: said longitudinal portion of said frame is said part of said carabiner secured within said through-bore of said cap.

8. The carabiner recited in claim 7, wherein said first length is 50-75% longer than said second length.

9. The carabiner recited in claim 7, wherein said first length is at least 66% longer than said second length.
